

Corporate Finance (in lingua inglese) - A.A. 2025/2026 - Topic 5e and 8: NPV and other investment criteria

TEXT and SOLUTION (through the use of Excel)

Vega is an international oil company. A previous investment of € 1,900 identified new and large deposits of crude oil in North Sea. Now Vega Corporation would like to buy a new equipment to extract it and the cost of the equipment would be € 2,500 and could be depreciated over a period of 5 years by the straight-line method. Project life: 5 years. The additional annual revenues would be € 1,000 and the extra costs (depreciation is not included) would be € 450, respectively, over the entire period. The working capital, throughout the period being considered, would be as follows (year-ends in parentheses): 250 (1), 550 (2), 750 (3), 600 (4) and 0 (5). No salvage value. The tax rate is 40%, the risk-free rate is 3%, the beta of the company is 2 and the expected market return is 8%. The cost of debt is 7%. The asset-to-equity ratio is 2. The risk associated with the project and its indebtedness are comparable to those of the firm as a whole. On the basis of the NPV and IRR rules, determine whether the investment proposal should be undertaken. Verify if the IRR you calculated is correct. Lastly, suppose the project will be financed only by the shareholders and their opportunity cost of capital is the same as before: determine the new NPV and IRR.

$$\text{WACC} = r_E \cdot E/V + r_D(1-t_c) \cdot D/V = (0,03 + 2 \cdot (0,08 - 0,03)) \cdot 0,5 + 0,07 \cdot (1 - 0,4) \cdot 0,5 = 0,086 = 8.6\%$$

		Period					
Year		0	1	2	3	4	5
Panel A Capital Investment							
1	Cash flow from investment in fixed assets	-2500,00					
2	Sales of fixed assets						
3	Less tax on sales						
4	Cash flow from capital investment (1+2+3)	-2500,00					
Panel B Operating Cash Flow							
5	Revenues		1000,00	1000,00	1000,00	1000,00	1000,00
6	Expenses*		450,00	450,00	450,00	450,00	450,00
7	Depreciation		500,00	500,00	500,00	500,00	500,00
8	Pretax profit (5 - 6 - 7)		50,00	50,00	50,00	50,00	50,00
9	Tax at 40%		20,00	20,00	20,00	20,00	20,00
10	Profit after tax (8 - 9)		30,00	30,00	30,00	30,00	30,00
11	Operating cash flow (10 + 7)		530,00	530,00	530,00	530,00	530,00

Panel C Investment in Working Capital							
12	Working capital		250,00	550,00	750,00	600,00	0,00
13	Change in working capital		250,00	300,00	200,00	-150,00	-600,00
14	Cash flow from investment in w. c . (-13)		-250,00	-300,00	-200,00	150,00	600,00

Panel D Project Valuation							
15	Total project cash flow (4 + 11 + 14)	-2500,00	280,00	230,00	330,00	680,00	1130,00
16	Discount factor at 8.6%	1,00	0,92	0,85	0,78	0,72	0,66
17	Discounted cash flows (15 x 16)	-2500,00	257,83	195,02	257,65	488,87	748,05
18	NPV (sum of discounted cash flows)	-552,60					

* Not including depreciation

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)

Calculations of the discounted cash flows at 8.6%						
Period	0	1	2	3	4	5
Formula	$-2500,00/(1+8.6\%)^0$	$280/(1+8.6\%)^1$	$230/(1+8.6\%)^2$	$330/(1+8.6\%)^3$	$680/(1+8.6\%)^4$	$1130/(1+8.6\%)^5$
Discounted cash flows	-2500,00	257,83	195,02	257,65	488,87	748,05

The proposal should be rejected as the NPV is less than 0.

IRR

IRR = 1,5463%

The proposal should be rejected as the IRR is lesser than the opportunity cost of capital, that is the WACC.

You may verify that if the IRR is correct, then the NPV at 1.5463% will be equal to 0:

Period	0	1	2	3	4	5
Formula	$-2500,00/(1+1.5463\%)^0$	$280/(1+1.5463\%)^1$	$230/(1+1.5463\%)^2$	$330/(1+1.5463\%)^3$	$680/(1+1.5463\%)^4$	$1130/(1+1.5463\%)^5$
Present values	- 2.500,00	275,74	223,05	315,15	639,52	1.046,55
Net present value (sum of PVs)	0,00					

$$r_e = 0,03 + 2 \cdot (0,08 - 0,03) = 0,13 = 13\%$$

		Period					
Year		0	1	2	3	4	5
Panel A Capital Investment							
1	Cash flow from investment in fixed assets	-2500,00					
2	Sales of fixed assets						
3	Less tax on sales						
4	Cash flow from capital investment (1+2+3)	-2500,00					

Panel B Operating Cash Flow							
5	Revenues		1000,00	1000,00	1000,00	1000,00	1000,00
6	Expenses*		450,00	450,00	450,00	450,00	450,00
7	Depreciation		500,00	500,00	500,00	500,00	500,00
8	Pretax profit (5 - 6 - 7)		50,00	50,00	50,00	50,00	50,00
9	Tax at 40%		20,00	20,00	20,00	20,00	20,00
10	Profit after tax (8 - 9)		30,00	30,00	30,00	30,00	30,00
11	Operating cash flow (10 + 7)		530,00	530,00	530,00	530,00	530,00

Panel C Investment in Working Capital							
12	Working capital		250,00	550,00	750,00	600,00	0,00
13	Change in working capital		250,00	300,00	200,00	-150,00	-600,00
14	Cash flow from investment in w. c . (-13)		-250,00	-300,00	-200,00	150,00	600,00

Panel D Project Valuation							
15	Total project cash flow (4 + 11 + 14)	-	2.500,00	280,00	230,00	330,00	1130,00
16	Discount factor at 13%		1,00	0,88	0,78	0,69	0,54
17	Discounted cash flows (15 x 16)	-	2.500,00	247,79	180,12	228,71	613,32
18	NPV (sum of discounted cash flows)		-813,01				

* Not including depreciation

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)

Calculations of the discounted cash flows at 13%						
Period	0	1	2	3	4	5
Formula	$-2500,00/(1+13\%)^0$	$280/(1+13\%)^1$	$230/(1+13\%)^2$	$330/(1+13\%)^3$	$680/(1+13\%)^4$	$1130/(1+13\%)^5$
Discounted cash flows	-2500,00	247,79	180,12	228,71	417,06	613,32

IRR

IRR = 1,5463%

The IRR does not change because the IRR only depends on the values of the cash flows.